

Hybrid Information Retrieval for Tip-of-the-Tongue Dataset

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Abstract—This study investigates the efficacy of lexical, dense, and hybrid models in dealing with the Tip-of-the-Tongue retrieval dataset. The BM25 model for lexical search, the SBERT model for dense search, and the hybrid approach, which combines retrieval models with neural re-ranking, are used in this study. The models were also evaluated using standard information retrieval metrics like Recall@K, MRR, and NDCG@K.

Keywords—BM25, SBERT, MonoT5, Recall@K, MRR, NDCG@K

I. INTRODUCTION

Information Retrieval (IR) systems aim to find relevant documents based on user queries. However, queries received by IR systems are often unclear and not well-defined. One such example is the Tip-of-the-Tongue (ToT) problem, in which users try to recall an item without using exact keywords. In such cases, vocabulary mismatch occurs, which affects the performance of lexical models, such as BM25. Recent studies [1] have shown the efficiency of using dense embedding models, such as SBERT (Sentence BERT), which handle unclear queries by incorporating semantic similarity. Recent work [2] suggests that the impact of MonoT5 is strong enough that the choice of base model (lexical or dense) has minimal effect on hybrid models, which we aim to verify in our experiment. Additionally, we aim to investigate whether the choice of base retriever affects hybrid performance. Therefore, we include combinations such as BM25 + SBERT, where lexical and dense scores are combined. The goal of this research is to examine the performance of the following retrieval models: BM25 (lexical), BM25 + MonoT5 (hybrid lexical), SBERT (dense), BM25 + SBERT, and SBERT + MonoT5 (hybrid dense). The models are evaluated using Recall@k, Mean Reciprocal Rank (MRR), and NDCG@k. The Recall@k is used to check if the correct answer is included within the top k results. MRR is used to measure the position at which the correct answer appears. NDCG@k is used to consider the position of the answer. We believe that semantic models will perform better than lexical models. Since semantic models capture context within ambiguous queries better than lexical models.

II. LITERATURE REVIEW

Traditional information retrieval systems rely on exact matching methods such as BM25. This study [3] explains that

BM25 is based on term frequency and inverse document frequency. However, BM25 depends on exact term matching and fails to capture semantic similarity. Since our problem mainly involves retrieving documents based on semantic similarity rather than exact matching, we investigate whether BM25 is an effective option. To overcome this limitation, dense retrieval models have been proposed [1]. Specifically, SBERT is selected as a dense retrieval model because it is effective in capturing semantic relationships between queries under vague query conditions, which makes it suitable for the ToT task. In addition to dense retrieval, a hybrid retrieval approach is used to improve overall performance. Recent studies [4] propose that hybrid models can improve retrieval effectiveness. Therefore, we incorporate MonoT5 as a re-ranking algorithm in our hybrid retrieval setups. This suggests that applying neural re-ranking on top of initial retrieval can significantly improve ranking quality. Furthermore, in this study, we investigate the findings of [2], which suggest that it remains unclear whether the choice of base model (lexical or dense) significantly affects hybrid performance when combined with strong re-ranking models such as MonoT5 (hybrid model). Therefore, we compare BM25, SBERT, and their hybrid extensions with MonoT5. We aim to better understand the differences between lexical and dense hybrid models by using this approach.

III. RESEARCH QUESTIONS AND METHODOLOGY

In this study, we seek to understand how well different information retrieval architectures fare in dealing with under-specified and vague queries, particularly those in the Tip-of-the-Tongue (ToT) dataset. These architectures include BM25, SBERT, BM25 + MonoT5, and SBERT + MonoT5. BM25 + SBERT is also included in this study to determine whether combining lexical and dense models without re-ranking improves performance. The research questions of this study are: (1) How significant is the performance of information retrieval models when MonoT5 is used as a re-ranker? (2) How significant is the choice of first-stage retriever, BM25 or SBERT, in determining the effectiveness of hybrid models that use MonoT5? (3) How does vagueness in query terms affect the relative performance of lexical, dense, and hybrid models? To evaluate all models, this study used standard information retrieval evaluation metrics, including Recall@k, Mean Reciprocal Rank, and Normalized Discounted Cumulative Gain.

BM25 uses term matching, whereas SBERT uses semantic vector space matching. For hybrid models, MonoT5 is used to re-rank models, whereas BM25 + SBERT uses a combination of normalization and weighted aggregation. We also perform exploratory data analysis (EDA) of the lengths of the queries and documents, term distributions, and vocabulary overlaps in order to further comprehend the impact of vagueness. This methodology enables us to make a systematic comparison of the lexical, dense, and hybrid approaches in order to identify the most effective approaches for the ToT task.

IV. DATA OVERVIEW

In this study, the TREC 2024 Tip-of-the-Tongue dataset is used for an information retrieval task. It contains 3 parts: queries, qrels, and a document corpus. The aim is to find the correct document from the corpus based on queries. However, these queries sometimes do not contain exact keywords; instead, they are unclear and vague. It has three splits: train, dev1, and dev2. For this study, the train split is used for training, dev1 is used for validation, and dev2 is used for testing. The queries split has 150 queries, divided into two columns: query_id and query. The query_id is the unique identifier, and query is the query string. In addition, the qrels split has four columns: query_id, iteration, doc_id, and relevance. The relevance column contains the relevance label, and all queries in the dataset are relevant to one document. Also, the corpus has 3,185,450 documents in five columns: doc_id, title, wikidata_id, text, and sections. The number of queries is small, but the corpus is very large. This is a challenging scenario in which the model has to select the right document from a very large corpus based on a vague description. The dataset was also examined for data quality. The dataset is clean and doesn't have any missing values or duplicate records. Hence, we don't have to perform any data cleaning. In summary, the TREC 2024 Tip of the Tongue dataset is a realistic and challenging dataset for retrieval.

V. EXPLORATORY DATA ANALYSIS

TABLE I. SUMMARY STATISTICS OF QUERIES

Split	Number of Queries	Number of Qrels	Avg. Query Length	Median Query Length
Train	150	150	147.38	127.0
Validation	150	150	139.33	118.5
Test	150	150	144.3	125.5

The number of queries and qrels is balanced for all splits. The average length of queries is relatively high, suggesting that queries are descriptive rather than keyword-based. The average and median lengths of queries are also close.

TABLE II. AVERAGE SEMANTIC SIMILARITY

Split	Average Similarity
Train	0.3577
Validation	0.3717
Test	0.3705

Semantic similarity between queries is computed using a pre-trained SBERT model (all-MiniLM-L6-v2). The average semantic cosine similarity values for all splits are moderate. Thus, there is a certain consistency in the data, but it is also diverse.

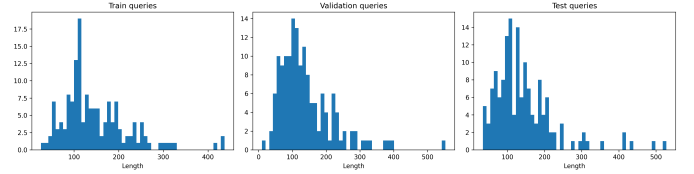


Fig. 1. Distribution of query lengths across train, validation, and test splits

Based on Fig. 1, their distribution is similar. Most queries range in medium length, but there are some longer queries. This suggests that queries tend to be descriptive rather than keyword-based. This could give semantic models such as SBERT and MonoT5 an advantage over lexical models such as BM25.

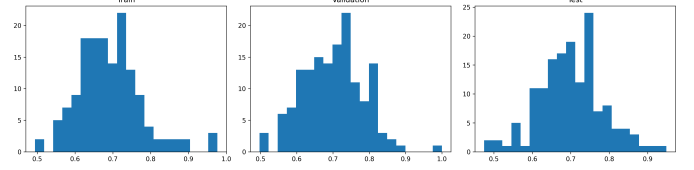


Fig. 2. Distribution of unique words across train, validation, and test splits

The relatively high unique-word ratio suggests that many queries are descriptive and lexically varied. This may make exact lexical matching harder and motivates the use of semantic retrieval models in addition to lexical baselines.

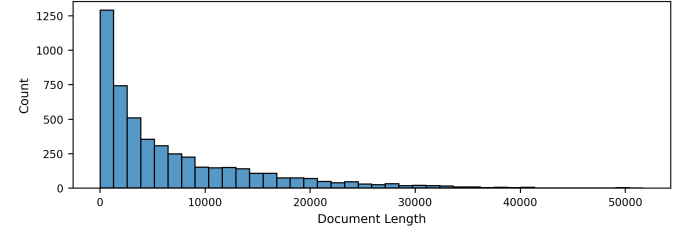


Fig. 3. Distribution of document lengths in the corpus.

Due to the large corpus size (3M+ documents), we analyze a sampled subset for efficiency. The distribution of corpus length is right-skewed. Most of the documents are of relatively short length, while some of the documents are of extremely long length. There are some outliers in the data.

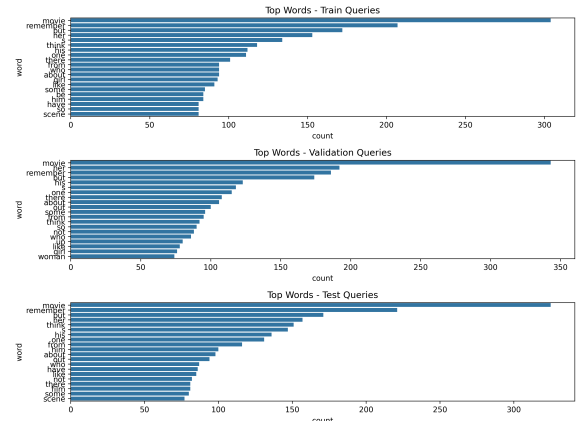


Fig. 4. Top 20 Open Class Words

The most frequent 20 words are similar across train, validation, and test sets. Words such as “movie”, “remember”, and “think” appear consistently across all splits. In particular, “movie” is the most frequent word; this shows us that the dataset is strongly focused on the movie domain.

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